

HRMS v2.0 Migration - TO DO List

Living document - updated as tasks complete. Source of truth for the migration's remaining work.

Milestone status

Completed tasks	16 / 22
Current branch	main @ 8ff66bc
Next task	Service-layer rewrite inc 3: PG re-run of extended probe (26 write flows) against clean hrms_probe, then fix any surfaced drift
Phase 0-1 progress	3 / 3 done
Phase 2 progress	2 / 2 done
Phase 3a progress	3 / 3 done
Phase 3b progress	8 / 9 done
Phase 4 progress	0 / 3 done
Phase 5 progress	0 / 1 done
Phase 6 progress	0 / 1 done

TO DO / status by phase

Phase 0-1 (3/3 done)

DONE	Freeze the v1.0 DuckDB schema; inventory tables for ETL Schema freeze note in docs/MIGRATION.md (08f7e40)
DONE	Build v2.0 target schema (db/postgres_schema.sql + Alembic baseline) 49-table public schema, CC-01..CC-16 documented (08f7e40)
DONE	One-time DuckDB -> PostgreSQL ETL with reconciliation hrms DB seeded; counts reconciled (08f7e40)

Phase 2 (2/2 done)

DONE	DuckDB->psycopy adapter (db_backend.py): translate/strftime/autocommit App runs on PostgreSQL legacy schema (b9164f2)
DONE	Unit + browser suites green on PostgreSQL (legacy schema preserved) 29/29 unit, 15/15 Playwright (b9164f2)

Phase 3a (3/3 done)

DONE	Argon2id password hashing; legacy bcrypt re-hashed on login security.py (98563ee)
DONE	Global CSRF enforcement (fetch wrapper + csrf_token field + token API) 98563ee
DONE	Server-side Redis sessions (REDIS_URL opt-in) + login rate limiting 98563ee

Phase 3b (8/9 done)

DONE	CC-01 identity rule enforced: scripts/check_cc_rules.py + PG-gated test 45 identity + 4 natural keys, sequences ahead of data (acb5219)
DONE	Public-flip readiness probe (scripts/probe_public_flip.py) GET + write-flow matrix against a throwaway public DB (acb5219)
DONE	Boolean adapter compat: predicates, INSERT params, naive datetime round-trip Inert on legacy (zero boolean cols); PG-gated tests (21480fe)

DONE	Write-flow probe + salary_structures seed data fix (CC-05) 11/11 core write flows green on public (21480fe)
DONE	CC-09 transactional outbox (outbox.py + scheduler + admin endpoints) Atomic business-write + event; backoff -> dead-letter (7e52f88)
DONE	CC-07 idempotency: @idempotent decorator + idempotency_keys wired for keyed POST retries Replay-without-duplicate, 409 on body reuse, claim released on failure; 6 unit tests green on every stack; probe replays on pure v2.0 JSONB
DONE	Service-layer rewrite inc 1: expanded audit_log (actor/entity/entity_id/before/after/request_id, CC-13) + notifications.category (FR-NOT-03) 8ff66bc; +6 unit tests; DuckDB 46 green
DONE	Service-layer rewrite inc 2: shift_assignments replaces users.shift_start/shift_end (FR-ATT-17); init_db no longer mutates v2.0 public.users get_shift/set_shift helpers reroute ~10 touch points; user CRUD + seed via set_shift; 7 new unit tests; DuckDB 53 green
IN PROGRESS	Extend probe write section: forgot-password, payroll bank-file/TDS, ticket/ATS + verify against a clean public schema Probe extended to 26 write flows in code; tickets/offer_letters/payroll_runs INSERTs made schema-explicit; PG re-run pending

Phase 4 (0/3 done)

PENDING	Attendance finalisation job (FR-JOB-01) SRS §14 Phase 4
PENDING	Maker-checker payroll (FR-PAY-06) SRS §14 Phase 4
PENDING	Corrected ATS / onboarding / offboarding flows SRS §14 Phase 4

Phase 5 (0/1 done)

PENDING	Final cutover: flip APP_DB_SCHEMA to public, retire legacy Out of the ETL blueprint; tracked in SRS
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Phase 6 (0/1 done)

PENDING	Decommission DuckDB runtime Out of the ETL blueprint; tracked in SRS
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Test & readiness gates

Gate	Stack	Current result
Unit suite (tests/test_app.py)	DuckDB	53 passed, 4 skipped (PG-gated CC-01/boolean/shift-public tests)
Unit suite (tests/test_app.py)	PostgreSQL	43 passed
Unit suite (tests/test_app.py)	PostgreSQL + Redis	43 passed
Browser suite (tests/test_playwright.py)	PostgreSQL	15 passed
CC-01 rule checker (scripts/check_cc_rules.py)	hrms (public)	OK - 45 identity + 4 natural keys
Public-flip probe (scripts/probe_public_flip.py)	hrms_probe (public)	86/86 GET + 12/12 write flows (measured pre-inc-2); write section extended to 26 flows - re-run pending

Update log (newest first)

2026-09-24	Service-layer rewrite inc 2 (shift_assignments): shift service helpers (get_shift/set_shift) replace ~10 direct users.shift_start/end touch points; init_db no longer mutates v2.0 public.users; user CRUD reroutes to shift_assignments; tickets/offer_letters/payroll_runs INSERTs made schema-explicit; DuckDB unit suite 53 green. Probe write section extended to 26 flows (forgot/reset pwd, shifts, tickets, ATS offer, payroll bank-file/TDS) — PG re-run pending.
2026-09-23	CC-07 idempotent writes done: @idempotent decorator on 9 POST routes, idempotency_keys DDL + hourly purge, 6 unit tests green (DuckDB 40 / PG 43 / PG+Redis 43), probe replays on pure v2.0 JSONB -> 86/86 GET + 12/12 write. Next: service-layer rewrite.
2026-09-23	TODO list created. Next task slated: CC-07 idempotency.

Generated 2026-09-24 by scripts/update_todo_pdf.py